

Def.

Let  $f: [a, b] \rightarrow \mathbb{C}$  be a function. Given a partition  $P = \{x_0, \dots, x_n\}$ , define

$$V(f, P) \stackrel{\text{def}}{=} \sum_{i=1}^n |f(x_i) - f(x_{i-1})|$$

which is called the variation of  $f$  with respect to a partition  $P$ . And, the value

$$TV(f) \stackrel{\text{def}}{=} \sup \{V(f, P) \mid P \text{ is partition on } [a, b]\}$$

is called the total variation of  $f$

If the total variation is finite, we say  $f$  is of bounded variation.

Prop.

If  $f: [a, b] \rightarrow \mathbb{C}$  is continuously differentiable, then  $f$  is of bounded variation with

$$TV(f) = \int_a^b |f'(t)| dt$$

Proof. Given a partition  $P = \{x_0, \dots, x_n\}$  on  $[a, b]$ , by the fundamental theorem of calculus,

$$V(f, P) = \sum_{i=1}^n |f(x_i) - f(x_{i-1})| = \sum_{i=1}^n \left| \int_{x_{i-1}}^{x_i} f'(t) dt \right| \leq \sum_{i=1}^n \int_{x_{i-1}}^{x_i} |f'(t)| dt = \int_a^b |f'(t)| dt$$

Since  $P$  was arbitrary,  $TV(f) \leq \int_a^b |f'(t)| dt$ , so the total variation has finite value.

To show that the reverse inequality, we use the fact that:

$f'$  is continuous uniformly, because it is continuous on the compact set  $[a, b]$ .

Given  $\varepsilon > 0$ , take  $\delta_1 > 0$  such that  $|x - t| < \delta_1 \Rightarrow |f'(x) - f'(t)| < \varepsilon$ .

And, take  $\delta_2 > 0$  such that if a partition  $P$  on  $[a, b]$  satisfies  $\|P\| < \delta_2$ , then

$$\left| \int_a^b |f'(t)| dt - \sum_{i=1}^n |f'(z_i)| \cdot |x_{i+1} - x_i| \right| < \varepsilon$$

where  $P = \{x_0, \dots, x_n\}$  and  $z_i \in [x_i, x_{i+1}]$  is arbitrary. The existence of  $\delta_2$  is given by the fact that  $|f'|$  is Riemann-integrable.

Prop. Let  $f: [a, b] \rightarrow \mathbb{R}$  be a function.  
Given  $c \in (a, b)$ ,

$$TV_a^b(f) = TV_a^c(f) + TV_c^b(f)$$

Proof. Given partitions  $P$  on  $[a, c]$  and  $Q$  on  $[c, b]$ , the union  $P \cup Q$  is a partition on  $[a, b]$ . Therefore, the equality given by definition

$$V_a^c(f, P) + V_c^b(f, Q) = V_a^b(f, P \cup Q) \leq TV_a^b(f)$$

holds. Since  $P$  and  $Q$  were arbitrary,

$$TV_a^c(f) + TV_c^b(f) \leq TV_a^b(f).$$

To obtain the reverse inequality, let  $R$  be a partition on  $[a, b]$ .  
Then,

$$\begin{aligned} V_a^b(f, R) &\leq V_a^b(f, R \cup \{c\}) = V_a^c(f, (R \cup \{c\}) \cap [a, c]) + V_c^b(f, (R \cup \{c\}) \cap [c, b]) \\ &\leq TV_a^c(f) + TV_c^b(f). \end{aligned}$$

Since  $R$  was arbitrary,

$$TV_a^b(f) \leq TV_a^c(f) + TV_c^b(f)$$

So, Proved. ~~Q~~

Directly,  $V(x) = TV_a^x(f)$  is well-defined monotone increasing function, if  $f \in BV$ .

Prop. Let  $f: [a, b] \rightarrow \mathbb{R}$  be a bounded variation.  
 Then, for any  $x \in [a, b]$ , the right-sided limit  $f(x^+)$  exists.  
 Similarly, for any  $x \in (a, b]$ , the left-sided limit  $f(x^-)$  exists.  
 Proof. Since  $f$  is of bounded variation, a map

$$V: [a, b] \rightarrow \mathbb{R} : t \mapsto TV_a^t(f)$$

is well-defined. Let  $x \in [a, b]$  be given. Then, given  $x < t_1 < t_2 \leq b$ ,

$$|f(t_2) - f(t_1)| \leq TV_{t_1}^{t_2}(f) = V(t_2) - V(t_1)$$

Let  $\{x_n\}$  be a sequence in  $[x, b]$  converging to  $x$ .  
 Since  $V$  is monotone increasing,  $V(x^+) = \inf\{V(t) \mid t > x\}$  exists,  
 therefore a sequence  $\{V(x_n)\}$  converges to  $V(x^+)$ .

Now, given  $\varepsilon > 0$ , there are  $N_1, N_2 \in \mathbb{N}$  such that  
 $n \geq N_1 \Rightarrow |V(x_n) - V(x^+)| < \frac{\varepsilon}{2}$  and  $m \geq N_2 \Rightarrow |V(x_m) - V(x^+)| < \frac{\varepsilon}{2}$   
 so  $m, n \geq \max\{N_1, N_2\}$  implies

$$|V(x_n) - V(x_m)| \leq |V(x_n) - V(x^+)| + |V(x^+) - V(x_m)| < \varepsilon$$

and since  $|f(x_n) - f(x_m)| \leq |V(x_n) - V(x_m)|$ ,  $\{f(x_n)\}$  is also a Cauchy sequence,  
 so the limit  $f(x^+)$  of  $\{f(x_n)\}$  exists. Since the sequence  $\{x_n\}$  in  $(x, b]$  was  
 arbitrary, the right-sided limit  $f(x^+)$  exists.

The left-sided limit exists by the same way.  $\square$

Note: But, the Converse is not true, let

$$f: [0, 1] \rightarrow \mathbb{R} : \begin{cases} 0 & \text{if } x=0 \\ x \cdot \sin \frac{1}{x} & \text{if } x \neq 0. \end{cases}$$

Then,  $f$  is not bounded variation, because: Consider the Partition on  $[0, 1]$

$$P_n = \left\{ 0, 1, \frac{2}{\pi}, \frac{2}{3\pi}, \dots, \frac{2}{(2n+1)\pi} \right\}$$

The variation  $V(f, P_n) \geq \frac{2}{\pi} \cdot \sum_{k=0}^{n-1} \frac{1}{2k+1}$  so,  $V(f, P_n) \rightarrow \infty$  as  $n \rightarrow \infty$ . But,

$$|f(x)| \leq \left| x \cdot \sin \frac{1}{x} \right| \leq |x| \rightarrow 0 \text{ as } x \rightarrow 0^+,$$

so the right-side limit exists.